

Stochastic and Information Thermodynamics

Course notes

Francisco Gómez Serrano

Contents

1	Information and entropy	3
1.1	Uncertainty	3
1.2	Information gained from a question	3
1.3	Codes	5
1.4	Mutual information	6
1.5	Relative entropy	7
2	Asymptotic equipartition and Shannon's theorems	7
2.1	Shannon's first theorem	7
2.2	Data compression	9
2.3	Shannon's second theorem: channel capacity	9
2.4	Sketch of the converse	11
3	Thermodynamics and information	11
3.1	Review of thermodynamics	11
3.2	Maxwell's demon	12
3.3	Szilard engine	12
3.4	Landauer–Bennett: the cost of erasure	13
3.5	Jarzynski equality	13
4	Markov processes and the master equation	14
4.1	Stochastic processes	14
4.2	Master equation	15
4.3	Detailed balance	16
4.4	Thermodynamics of the master equation	16
4.5	Several reservoirs: non-equilibrium steady states	18
5	Stochastic thermodynamics and fluctuation theorems	18
5.1	Thermodynamics along a trajectory	18

5.2	Fluctuation theorems	20
5.3	Measurement and feedback	20

1 Information and entropy

Information theory quantifies the *amount* of information contained in a message; it says nothing about its *meaning*. A random text carries more information than a predictable one, precisely because it cannot be predicted. Information is measured in **bits**.

1.1 Uncertainty

Alice and Bob both know a random variable X that takes the values $x_1, \dots, x_N$ with probabilities $p_1, \dots, p_N$ ($\sum_i p_i = 1$). Alice performs the experiment; Bob's **uncertainty** about its outcome is the Shannon entropy

$$H(X) := - \sum_i p_i \log_2 p_i \quad (\text{in bits}),$$

with the convention $0 \log 0 = 0$. Clearly $H(X) \geq 0$. Changing the base of the logarithm changes the unit:

$$1 \text{ bit} = \ln 2 \text{ nats}.$$

Example (Guessing a number). Guess the number I have in mind, among $0, 1, \dots, 7$, by asking yes/no questions. Bisecting the set at each step reduces the possibilities as $2^3 \rightarrow 2^2 \rightarrow 2^1 \rightarrow 2^0$. Indeed

$$H(X) = -8 \left[\frac{1}{8} \log_2 \frac{1}{8} \right] = \log_2 2^3 = 3 \text{ bits},$$

which is the minimum number of questions needed, on average, over a large number of games.

1.2 Information gained from a question

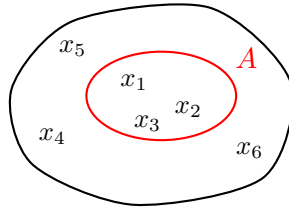


Figure 1.1: The question “is $x_i \in A$?” splits the outcomes into two sets.

We ask whether the outcome lies in a subset A :

$$\text{Prob}[\text{yes}] = p = \sum_{i \in A} p_i, \quad \text{Prob}[\text{no}] = 1 - p.$$

After the answer the probabilities are updated:

$$\text{yes: } p_i \rightarrow p'_i = \begin{cases} p_i/p, & i \in A, \\ 0, & i \notin A, \end{cases} \quad \text{no: } p_i \rightarrow p''_i = \begin{cases} 0, & i \in A, \\ p_i/(1-p), & i \notin A. \end{cases}$$

The remaining uncertainties are

$$H_{\text{yes}} = - \sum_{i \in A} \frac{p_i}{p} \log_2 \frac{p_i}{p} = - \frac{1}{p} \sum_{i \in A} p_i \log_2 p_i + \log_2 p,$$

$$H_{\text{no}} = - \frac{1}{1-p} \sum_{i \notin A} p_i \log_2 p_i + \log_2(1-p).$$

The mean final uncertainty (after playing many times) is

$$\begin{aligned} H_{\text{fin}} &= p H_{\text{yes}} + (1 - p) H_{\text{no}} = - \sum_i p_i \log_2 p_i + p \log_2 p + (1 - p) \log_2 (1 - p) \\ &= H_{\text{in}} + p \log_2 p + (1 - p) \log_2 (1 - p). \end{aligned}$$

The **information** gained is the reduction in uncertainty,

$$I := H_{\text{in}} - H_{\text{fin}} = -p \log_2 p - (1 - p) \log_2 (1 - p) =: h(p).$$

The answer itself is a random variable $Y \in \{\text{yes}, \text{no}\}$ with probabilities $p, 1 - p$, so $H(Y) = h(p)$ and

$$I = H(Y) \quad (\text{if the answer has no errors}) :$$

the information gained equals the uncertainty one had about the answer. The binary entropy $h(p)$ is maximal, $h = 1$ bit, for $p = \frac{1}{2}$: splitting into two equiprobable sets gives the largest possible information.

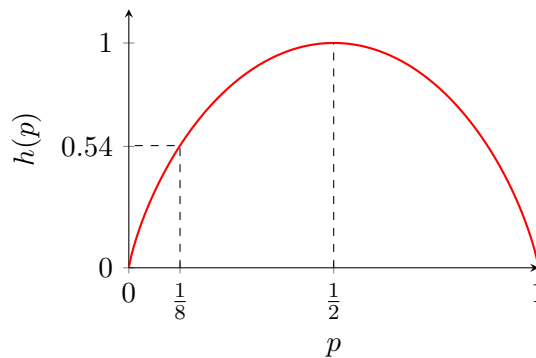


Figure 1.2: Binary entropy $h(p)$.

For instance, in the number-guessing game, a question that isolates a single number out of eight ($p = \frac{1}{8}$) yields only $h(1/8) \approx 0.54$ bits, so on average about $3/0.54 \approx 5.5$ such questions would be needed.

Exercise (Twelve coins). There are 12 coins, one of which is defective. With a two-pan balance one has to find the defective coin and decide whether it is lighter or heavier. How many weighings are needed?

There are 24 possible results $(i, \pm)$, $i = 1, \dots, 12$, so $H_{\text{in}} = \log_2 24 \approx 4.58$ bits. Each weighing has three outcomes (left down, right down, balanced), so it gives at most $H(Y) = \log_2 3 \approx 1.58$ bits. Hence at least $4.58/1.58 \approx 2.9$, i.e. three weighings are needed. The following protocol achieves it.

Solution. Notation: $\circ$ is a coin known to be genuine, “+” a coin that can only be defective by being heavier, “−” one that can only be lighter.

First weighing. Split the coins into three groups of four and put two of the groups on the balance.

- **Case 1: the balance stays level** (probability $1/3$). The defective coin is among the four coins left out, which are all suspects (either way). Take two of them to group I, one suspect plus one genuine coin $\circ$ to group II, and set the fourth suspect aside. *Second weighing:* I against II.
 - *1A: level.* The defective coin is the one set aside; weighing it against a genuine coin (*third weighing*) tells whether it is heavier or lighter.

- *1B: the balance tips*, say pan I goes down. Then either one of the two coins in I is heavy (+) or the suspect in II is light (–). *Third weighing*: one + coin together with the – coin against two genuine coins, keeping the other + aside. Down: the + coin is heavy; up: the – coin is light; level: the coin set aside is heavy. If pan I goes up in the second weighing the procedure is the same with the roles of + and – exchanged.
- **Case 2: the balance tips** (probability 2/3). The four coins in the lower pan become +, the four in the upper pan –, and the four left out are genuine. Form the groups

$$\text{I: } + + - \quad \text{II: } + - \circ \quad \text{III (aside): } + - -$$

Second weighing: I against II.

- *2A: level*. The defective coin is in III. *Third weighing*: + – of group III against $\circ \circ$, keeping the other – aside. Down: the + is heavy; up: that – is light; level: the – set aside is light.
- *2B: pan I goes down*. The suspects are the two + in I and the – in II. *Third weighing*: one + of I and the – of II against $\circ \circ$, keeping the other + aside; the result is read as before.
- *2C: pan I goes up*. The suspects are the – in I and the + in II. *Third weighing*: the – of I against $\circ$. Up: it is light; level: the + in II is heavy.

1.3 Codes

Bob’s uncertainty H is the minimum number of binary answers needed to know the result and therefore the number of bits Alice has to send to transmit it.

Example. To guess a number among $0, \dots, 7$ we can always ask the same questions: “> 3?”, “> 5?”, “> 6?”. For 6 the answers are yes, yes, no, i.e. 110. The answers to a fixed set of questions define a code:

number	0	1	2	3	4	5	6	7
code	000	001	010	011	100	101	110	111

This is just the **binary code**: $x \rightarrow (x_n, x_{n-1}, \dots, x_1, x_0)$ with $x = x_n 2^n + x_{n-1} 2^{n-1} + \dots + 2x_1 + x_0$. So that the receiver does not get confused, all codewords have the same length (3 bits).

Example (Horse race). Eight horses have winning probabilities and codewords

p_i	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{16}$	$\frac{1}{64}$	$\frac{1}{64}$	$\frac{1}{64}$	$\frac{1}{64}$
code	0	10	110	1110	111100	111101	111110	111111

The entropy is

$$\begin{aligned}
 H &= -\left(\frac{1}{2} \log_2 \frac{1}{2} + \frac{1}{4} \log_2 \frac{1}{4} + \frac{1}{8} \log_2 \frac{1}{8} + \frac{1}{16} \log_2 \frac{1}{16} + \frac{4}{64} \log_2 \frac{1}{64}\right) \\
 &= \frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{4}{16} + \frac{4 \cdot 6}{64} = 2 \text{ bits.}
 \end{aligned}$$

A sequence of results such as 001011101... can be decoded unambiguously (no codeword is a prefix of another), and the mean codeword length is

$$\langle L \rangle = \sum_i l_i p_i = 1 \cdot \frac{1}{2} + 2 \cdot \frac{1}{4} + \dots = 2 = H.$$

The lengths are $l_i = -\log_2 p_i$, so $\langle L \rangle = \sum_i (-\log_2 p_i) p_i = H(X)$.

For eight equiprobable outcomes ($H = 3$ bits) the 3-bit binary code is optimal. If the probabilities are not equal (say $H = 2.8$ bits) one can do better with variable-length codes. In general, for a random variable X with values $x_1, \dots, x_n$ and probabilities $p_1, \dots, p_n$, let $c(x_i)$ be the codeword assigned to each symbol and $l_c(x_i)$ its length. The optimal code satisfies

$$H(X) \leq \langle L \rangle = \sum_i l_c(x_i) p_i < H(X) + 1 \quad (\text{in bits}).$$

Block coding. Let $X = (z_1, \dots, z_m)$ be a block of m symbols, so each value $x_i = (z_1^{(i)}, \dots, z_m^{(i)})$ receives a codeword $c(x_i)$. Then

$$H(X) \leq \sum_i l_c(z_1^{(i)}, \dots, z_m^{(i)}) p_i < H(X) + 1.$$

If the z_j are i.i.d. (independent and identically distributed), $H(X) = mH(z)$ and

$$H(z) \leq \underbrace{\frac{1}{m} \sum_i l_c(z_1^{(i)}, \dots, z_m^{(i)}) p_i}_{\text{expected code length per symbol}} < H(z) + \frac{1}{m},$$

where p_i is now a product of probabilities. When $m \rightarrow \infty$ the expected code length per symbol equals $H(z)$.

Sequences. Toss a coin N times with outcomes 0, 1 of probabilities p and $q = 1 - p$. A particular sequence such as 000101110... with n heads and $N - n$ tails has probability $p^n q^{N-n}$. If N is very large, the law of large numbers says that the typical value is $n = Np$, so the probability of a typical sequence is

$$p^{Np} (1-p)^{N-Np} = 2^{N[p \log_2 p + (1-p) \log_2 (1-p)]} = 2^{-Nh(p)},$$

controlled by the entropy. This idea is made precise in Section 2.

1.4 Mutual information

For a question about X (“is $x_i \in A$?”, $p = \text{Prob}(x \in A)$) we found $H_{\text{fin}} + I = H_{\text{in}}$, with $I = h(p) = H(Y)$ when there is no error in the answer. In information theory, “information” means **mutual information**:

$$I(X : Y) := H(X) - H(X|Y),$$

where $H(X|Y)$ is the **conditional entropy**

$$H(X|Y) := \sum_i p(y_i) H(X|y_i), \quad H(X|y_i) = - \sum_j p(x_j|y_i) \log_2 p(x_j|y_i).$$

Using $p(y_i)p(x_j|y_i) = p(x_j, y_i)$,

$$\begin{aligned} I(X : Y) &= - \sum_j p(x_j) \log_2 p(x_j) + \sum_{i,j} p(x_j, y_i) \log_2 \frac{p(x_j, y_i)}{p(y_i)} \\ &= - \sum_j p(x_j) \log_2 p(x_j) - \sum_i p(y_i) \log_2 p(y_i) + \sum_{i,j} p(x_j, y_i) \log_2 p(x_j, y_i), \end{aligned}$$

that is,

$$I(X : Y) = H(X) + H(Y) - H(X, Y) = I(Y : X),$$

which is symmetric under $X \leftrightarrow Y$. Therefore

$$I(X : Y) = H(X) - H(X|Y) = H(Y) - H(Y|X).$$

For the yes/no question,

$$H(Y|X) = \sum_j p(x_j) H(Y|x_j), \quad p(y|x_j) = \begin{cases} 1 & y = \text{yes}, x_j \in A \text{ or } y = \text{no}, x_j \notin A, \\ 0 & \text{otherwise.} \end{cases}$$

Then $H(Y|x_j) = 0$ if the answer has no errors, and $I(X : Y) = H(Y)$: “the information equals the entropy”.

1.5 Relative entropy

Writing the mutual information as

$$I(X : Y) = \sum_{i,j} p(x_j, y_i) \log_2 \frac{p(x_j, y_i)}{p(x_j)p(y_i)}$$

suggests the following definition.

Definition. The **relative entropy** or **Kullback–Leibler divergence** between two distributions $p_1, \dots, p_n$ and $q_1, \dots, q_n$ over the same outcomes is

$$D(p||q) := \sum_i p_i \log \frac{p_i}{q_i} \geq 0.$$

It is sometimes called a “distance”, but it is not a metric (it is not symmetric).

$D(p||q) \geq 0$ follows from the convexity of $f(t) = -\ln t$ and Jensen’s inequality, $\sum_i \alpha_i f(t_i) \geq f(\sum_i \alpha_i t_i)$ for $\alpha_i \geq 0$, $\sum_i \alpha_i = 1$:

$$-\sum_i \alpha_i \ln t_i \geq -\ln \left(\sum_i \alpha_i t_i \right).$$

Taking $\alpha_i = p_i$ and $t_i = q_i/p_i$ (with $\sum_i q_i = 1$, $q_i \geq 0$),

$$\sum_i p_i \ln \frac{p_i}{q_i} \geq -\ln \left(\sum_i p_i \frac{q_i}{p_i} \right) = -\ln 1 = 0.$$

Hence

$$I(X : Y) = D(p(x, y) || p(x)p(y)) \geq 0.$$

Corollary. $I(X : Y) = 0$ if and only if X and Y are independent random variables, in which case $H(X, Y) = H(X) + H(Y)$.

Remark. The information obtained from a particular outcome, $I_i := H(X) - H(X|y_i)$, can be negative. Only its average is guaranteed to be non-negative: $\sum_i p(y_i) I_i = H(X) - H(X|Y) = I(X : Y) \geq 0$.

2 Asymptotic equipartition and Shannon’s theorems

2.1 Shannon’s first theorem

We give one version of Shannon’s first (source coding) theorem. The main tool is the **asymptotic equipartition property** (AEP), which is the law of large numbers written in a form useful for information theory.

Law of large numbers. Let $X_1, \dots, X_n$ be i.i.d. random variables (values x with probability $p(x)$) and $E(X)$ the expectation value of each X_i . Then

$$\frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{n \rightarrow \infty} E(X) \quad (\text{limit in probability}),$$

i.e. $\text{Prob} \left[\left| \frac{1}{n} \sum_{i=1}^n X_i - E(X) \right| < \varepsilon \right] \rightarrow 1$ as $n \rightarrow \infty$; equivalently, for all $\delta, \varepsilon > 0$ there is n_0 such that

$$\text{Prob} \left[\left| \frac{1}{n} \sum_{i=1}^n X_i - E(X) \right| < \varepsilon \right] > 1 - \delta \quad \text{for } n > n_0.$$

Note that if X is a random variable, so is any $f(X)$.

AEP.

$$-\frac{1}{n} \log_2 p(X_1, \dots, X_n) \xrightarrow{n \rightarrow \infty} H(X), \quad H(X) = - \sum_x p(x) \log_2 p(x) = E(-\log_2 p(X)).$$

Since the X_i are i.i.d., $p(x_1, \dots, x_n) = \prod_{i=1}^n p(x_i)$, so

$$-\frac{1}{n} \log_2 p(x_1, \dots, x_n) = \frac{1}{n} \sum_{i=1}^n (-\log_2 p(x_i)),$$

and applying the law of large numbers to the i.i.d. random variables

$$-\log_2 p(X_1), \dots, -\log_2 p(X_n)$$

gives the AEP.

Definition. The **typical set** $A_\varepsilon^{(n)}$ with respect to $p(x)$ is the set of sequences $(x_1, \dots, x_n)$, $x_i \in \mathcal{X}$ (the alphabet), such that

$$2^{-n(H(X)+\varepsilon)} < p(x_1, \dots, x_n) < 2^{-n(H(X)-\varepsilon)}.$$

As a consequence of the AEP:

- (i) If $(x_1, \dots, x_n) \in A_\varepsilon^{(n)}$, then $H(X) - \varepsilon \leq -\frac{1}{n} \log_2 p(x_1, \dots, x_n) \leq H(X) + \varepsilon$.
- (ii) $\text{Prob} \left[(x_1, \dots, x_n) \in A_\varepsilon^{(n)} \right] > 1 - \varepsilon$ for n large enough: almost all sequences that occur are typical.
- (iii) $|A_\varepsilon^{(n)}| \leq 2^{n(H(X)+\varepsilon)}$.
- (iv) $|A_\varepsilon^{(n)}| \geq (1 - \varepsilon) 2^{n(H(X)-\varepsilon)}$ for n large enough.

Proof. (i) By definition. (ii) We have

$$\text{Prob}[(x_1, \dots, x_n) \in A_\varepsilon^{(n)}] = \text{Prob} \left[\left| -\frac{1}{n} \log_2 p(x_1, \dots, x_n) - H(X) \right| < \varepsilon \right],$$

and by the AEP, for all ε, δ there is n_0 such that this probability exceeds $1 - \delta$ for $n > n_0$; we can take $\delta = \varepsilon$. (iii) Let $\mathbf{x}$ denote a sequence in $\mathcal{X}^n$ (of length n ; $|\mathcal{X}^n| = |\mathcal{X}|^n$, with $|\mathcal{X}|$ the cardinality of the alphabet). Then

$$1 = \sum_{\mathbf{x} \in \mathcal{X}^n} p(\mathbf{x}) \geq \sum_{\mathbf{x} \in A_\varepsilon^{(n)}} p(\mathbf{x}) \geq \sum_{\mathbf{x} \in A_\varepsilon^{(n)}} 2^{-n(H(X)+\varepsilon)} = 2^{-n(H(X)+\varepsilon)} |A_\varepsilon^{(n)}|,$$

with $|A_\varepsilon^{(n)}| = \sum_{\mathbf{x} \in A_\varepsilon^{(n)}} 1$. (iv) From (ii),

$$1 - \varepsilon < \sum_{\mathbf{x} \in A_\varepsilon^{(n)}} p(\mathbf{x}) \leq \sum_{\mathbf{x} \in A_\varepsilon^{(n)}} 2^{-n(H(X)-\varepsilon)} = 2^{-n(H(X)-\varepsilon)} |A_\varepsilon^{(n)}| \Rightarrow |A_\varepsilon^{(n)}| \geq (1 - \varepsilon) 2^{n(H(X)-\varepsilon)}.$$

2.2 Data compression

Data compression is a consequence of the AEP. Let $X_1, \dots, X_n$ be i.i.d. $\sim p(x)$.

- There are $|\mathcal{X}|^n$ sequences; in principle we need $n \log_2 |\mathcal{X}|$ bits to identify one of them.
- Divide the sequences into two sets: those in $A_\varepsilon^{(n)}$ and those in its complement $A_\varepsilon^{(n)c}$. Label the $|A_\varepsilon^{(n)}|$ typical sequences $1, 2, \dots, |A_\varepsilon^{(n)}|$; since $|A_\varepsilon^{(n)}| \leq 2^{n(H+\varepsilon)}$, the labels require at most $n(H+\varepsilon) + 1$ bits. Add a prefix 0 to indicate a typical sequence: typical sequences use at most $n(H+\varepsilon) + 2$ bits. This mapping is the key step.
- Similarly, the sequences in $A_\varepsilon^{(n)c}$ can be labelled with fewer than $n \log_2 |\mathcal{X}| + 1$ bits; with a prefix 1 they use at most $n \log_2 |\mathcal{X}| + 2$ bits.

Let $x^n := (x_1, \dots, x_n)$, $x_i \in \mathcal{X}$, and $l(x^n)$ the length (in bits) of its codeword. If n is large ($\text{Prob}[x^n \in A_\varepsilon^{(n)}] > 1 - \varepsilon$),

$$\begin{aligned} \mathbb{E}(l(x^n)) &= \sum_{x^n \in A_\varepsilon^{(n)}} l(x^n) p(x^n) + \sum_{x^n \in A_\varepsilon^{(n)c}} l(x^n) p(x^n) \\ &\leq [n(H+\varepsilon) + 2] \text{Prob}[A_\varepsilon^{(n)}] + [n \log_2 |\mathcal{X}| + 2] \text{Prob}[A_\varepsilon^{(n)c}] \\ &\leq n(H+\varepsilon) + 2 + \varepsilon(n \log_2 |\mathcal{X}| + 2) = n(H+\varepsilon'), \end{aligned}$$

with

$$\varepsilon' := \varepsilon + \varepsilon \log_2 |\mathcal{X}| + \frac{2(1+\varepsilon)}{n},$$

which can be made as small as we want by choosing ε small and n large. Since no code can do better than the entropy,

$$H(X) \leq \frac{\mathbb{E}(l(x^n))}{n} \leq H(X) + \varepsilon'.$$

The mean number of bits per symbol is the entropy.

2.3 Shannon's second theorem: channel capacity

Alice communicates with Bob through a physical medium, the **channel**. She sends $x \in \mathcal{X}$ and Bob receives $y \in \mathcal{Y}$, with transition probabilities $p(y|x)$. In a perfect channel each input is mapped to a definite output ($x_1 \rightarrow y_1, x_2 \rightarrow y_2$: a dictionary); in a noisy channel one input can produce several outputs ($x_2 \rightarrow y_1$ or y_2).

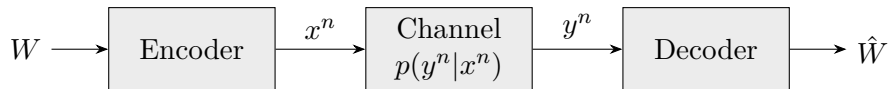


Figure 2.1: Communication scheme.

A message W is one of M equiprobable symbols $1, 2, \dots, M$. The encoder generates an **(M, n) code**: for each symbol a codeword $x^n = (x_1, \dots, x_n)$ of length n , where n is the number of uses of the channel. Bob receives $y^n = (y_1, \dots, y_n)$ and decodes it into $\hat{W}$. For a memoryless channel $p(y^n|x^n) = \prod_{i=1}^n p(y_i|x_i)$.

Definition. The **rate** of an (M, n) code is

$$R := \frac{\log_2 M}{n} \quad (\text{bits per channel use}).$$

$\log_2 M$ is the optimal length of the message, and n the length actually used. In a perfect binary channel, $R = 1$ (no redundancy).

Theorem (Shannon). The message can be transmitted with negligible error if and only if $R \leq C$, where C is the **channel capacity**

$$C := \max_{p(x)} I(X : Y), \quad I(X : Y) = D(p(x, y) \| p(x)p(y)) \geq 0.$$

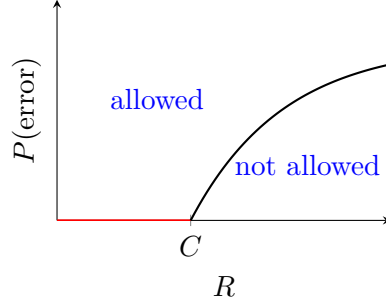


Figure 2.2: Achievable region: below capacity the error can be made arbitrarily small.

Example (Noiseless binary channel). $0 \rightarrow 0$ and $1 \rightarrow 1$ with certainty: $p(y|0) = \delta_{y,0}$, $p(y|1) = \delta_{y,1}$. With $M = 2$, $\log_2 M = 1$ and $R = 1$ if $n = 1$. Now

$$I(X : Y) = H(Y) - H(Y|X), \quad H(Y|X) = \sum_{x \in \{0,1\}} p(x)H(Y|x), \quad H(Y|0) = H(Y|1) = 0,$$

and $p(y) = \sum_x p(y|x)p(x)$ gives $p(y = 0) = p(x = 0)$, $p(y = 1) = p(x = 1)$, so $I(X : Y) = H(Y) = H(X)$. The maximum is attained for $p(x) = \{\frac{1}{2}, \frac{1}{2}\}$:

$$C = 1.$$

The $(2, 2)$ code $1 \rightarrow 00$, $2 \rightarrow 11$ has $R = \frac{1}{2}$.

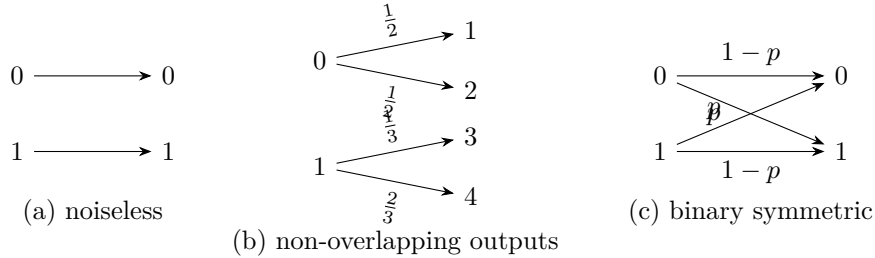


Figure 2.3: Examples of channels.

Example (Noisy channel with non-overlapping outputs). Input 0 produces 1 or 2 with probability $\frac{1}{2}$ each; input 1 produces 3 with probability $\frac{1}{3}$ and 4 with probability $\frac{2}{3}$. With $M = 2$ and $n = 1$, $R = 1$ bit per channel use. Then

$$\begin{aligned} H(Y|x=0) &= \log_2 2 = 1, \\ H(Y|x=1) &= -\frac{1}{3} \log_2 \frac{1}{3} - \frac{2}{3} \log_2 \frac{2}{3} = \log_2 3 - \frac{2}{3}, \\ H(Y|X) &= p(x=0) \cdot 1 + p(x=1) \left(\log_2 3 - \frac{2}{3} \right). \end{aligned}$$

With $p := p(x=0)$, $q := p(x=1)$: $p(y=1) = p(y=2) = p/2$, $p(y=3) = q/3$, $p(y=4) = 2q/3$, and

$$H(Y) = -p \log_2 \frac{p}{2} - \frac{q}{3} \log_2 \frac{q}{3} - \frac{2q}{3} \log_2 \frac{2q}{3} = h(p) + p + q \left(\log_2 3 - \frac{2}{3} \right).$$

Therefore $I(X : Y) = H(Y) - H(Y|X) = h(p)$, maximal for $p = q = \frac{1}{2}$: $C = 1$. Since the outputs never overlap, the noise does not reduce the capacity.

Example (Binary symmetric channel). Each bit is flipped with probability p : $p(0|0) = p(1|1) = 1 - p$, $p(1|0) = p(0|1) = p$. Then $H(Y|X) = h(p)$, and $H(Y) \leq 1$ with equality for uniform input, so

$$C = 1 - h(p) < 1 \quad \text{if } p \neq 0.$$

We can still send information, but we need more uses of the channel. With $M = 2$, $R = \frac{\log_2 2}{n} = \frac{1}{n} \leq 1 - h(p)$. For $p = \frac{1}{8}$, $h(p) \approx 0.54$ and

$$n \geq \frac{1}{1 - h(1/8)} \approx \frac{1}{0.46} \approx 2.2,$$

so about 3 bits per symbol are needed.

Repetition code. With $n = 1$, the bit 0 is received wrongly with probability p . With $n = 3$, encode $0 \rightarrow 000$ and decode by majority vote. The decoding fails if two or three bits are flipped (111 with probability p^3 ; 110, 101, 011 with probability $p^2(1 - p)$ each):

$$p_{\text{error}} = p^3 + 3p^2(1 - p) < p \quad (p < \frac{1}{2}).$$

2.4 Sketch of the converse

Consider an (M, n) code: messages $W \in \{1, \dots, M\}$, each mapped to $x_W^n = (x_1, \dots, x_n)$; the channel is a physical object characterized by $p(y^n|x^n) = \prod_{i=1}^n p(y_i|x_i)$. There exist (M, n) codes with rate $R \leq C$, i.e. with $M = 2^{nR}$ messages, that can be decoded without error, $\hat{W} = W$.

Example. $W \in \{A, B, C, D\}$ encoded as 00, 01, 10, 11: $M = 4$, $\log_2 M = 2$ and $R = 2/n$. Through the noiseless binary channel ($C = 1$) we need $n = 2$, and $R = 1 \leq C$.

Each input x^n can be mapped by the channel to several outputs y^n . For typical input sequences the outputs are also typical: for fixed n , there are about $2^{nH(Y|X)}$ typical outputs for each input, while the total number of typical output sequences is $2^{nH(Y)}$. For the M codewords to be distinguishable, their sets of outputs must not overlap:

$$\begin{aligned} M 2^{nH(Y|X)} &\leq 2^{nH(Y)} \Rightarrow \log_2 M + nH(Y|X) \leq nH(Y) \\ \Rightarrow R &= \frac{\log_2 M}{n} \leq H(Y) - H(Y|X) = I(X : Y). \end{aligned}$$

Therefore $R \leq \max_{p(x)} I(X : Y) = C$.

3 Thermodynamics and information

3.1 Review of thermodynamics

- **First law:** $\Delta E = W + Q$, with W the work done on the system and Q the heat absorbed by it.
- **Second law:** the entropy production (change of entropy of the universe), with the environment a bath at temperature T ,

$$\Sigma = \Delta S + \left(-\frac{Q}{T}\right) \geq 0.$$

3.2 Maxwell's demon

A gas at temperature T fills a box with adiabatic walls, divided in two by a wall with a small door. A demon watches the molecules and opens the door only to let fast molecules ($v > \bar{v}$) go to the right and slow ones ($v < \bar{v}$) go to the left. The velocities follow the Maxwell distribution (the demon is “Maxwellian”). After a while the left side is colder and the right side hotter, without any work: $\Sigma < 0$, an apparent violation of the second law (paradox).

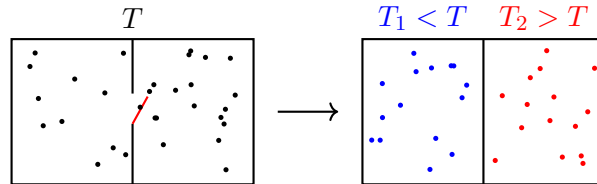


Figure 3.1: Maxwell's demon sorts fast and slow molecules: apparently $\Sigma < 0$.

3.3 Szilard engine

Consider a single particle ($N = 1$) in a box of volume V in contact with a bath at temperature T , so that $pV = k_B T$ and $E = \frac{3}{2} k_B T$.

1. Insert a partition in the middle and measure on which side (L or R) the particle is: this is 1 bit of information (1 bit = $\ln 2$ nats).
2. Knowing the side, turn the partition into a piston and let the gas expand reversibly and isothermally back to the full volume:

$$W = - \int_{V/2}^V p dV' = -k_B T \int_{V/2}^V \frac{dV'}{V'} = -k_B T \ln 2 < 0 :$$

the gas has done work on the agent.

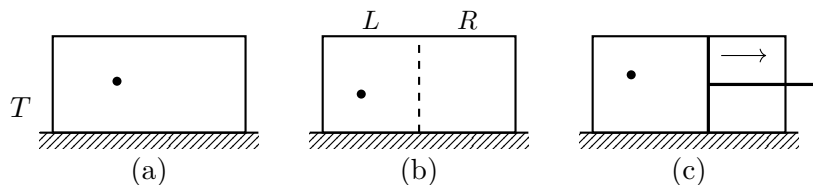


Figure 3.2: Szilard engine: (a) one particle in a box; (b) a partition is inserted and the side is measured; (c) the partition acts as a piston and the gas expands.

After one cycle the system returns to its initial state ($\Delta E = 0$, $\Delta S = 0$): the energy is a function of T only, so

$$\Delta E = 0 = W + Q \Rightarrow Q = -W = k_B T \ln 2$$

has been extracted from a single heat bath and converted to work. This contradicts the Kelvin–Planck statement (“there is no process whose only effect is to extract energy from a heat bath and convert it into work”). The entropy production in the cycle is

$$\Sigma = \Delta S + \left(-\frac{Q}{T} \right) = -k_B \ln 2 < 0,$$

which contradicts the second law.

Brillouin. There is an energy cost to measure, such that $W_{\text{meas}} + \bar{W} \geq 0$ with $\bar{W} = -k_B T \ln 2$. However, one can always measure more precisely, and it is not clear whether there is a limit to the efficiency.

Szilard. The demon is itself a thermodynamic system. Its memory D records the result of each measurement. Before the measurements every memory is in the standard state ($D = 0$), with entropy $S(M) = H(M) = 0$. After them, $D = 0$ (L) or $D = 1$ (R) with equal probabilities, so the demon's entropy has increased to

$$S(M) = k_B H(M) = k_B \ln 2$$

(H in nats, entropy in units of k_B). Including the memory,

$$\Sigma_{\text{Szilard}} = \Delta S + \left(-\frac{Q}{T}\right) + \Delta S(M) = -k_B \ln 2 + k_B \ln 2 = 0.$$

Even so, the Kelvin–Planck statement is still violated: the cycle is not closed, since the memory has not returned to its initial state.

3.4 Landauer–Bennett: the cost of erasure

The memory of the demon has to be erased to close the cycle, and the cost is

$$W_{\text{erase}} \geq k_B T \ln 2,$$

which follows from thermodynamics. The argument: to erase the memory we need a protocol that is *independent* of the state of the memory.

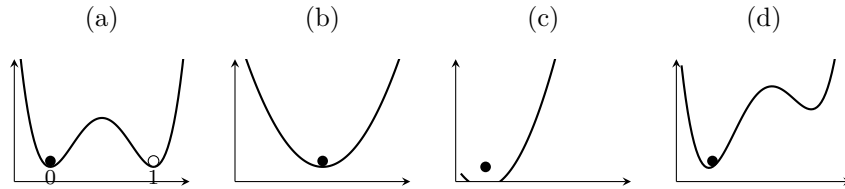


Figure 3.3: Erasure protocol for a one-bit memory (a particle in a double well): (a) the bit is 0 or 1; (b) the barrier is lowered; (c) the potential is tilted so that the particle ends up on the left; (d) the barrier is restored, and the memory is in state 0 regardless of its initial state.

For the process, $S_{\text{in}} = k_B \ln 2$, $S_{\text{fin}} = 0$, so $\Delta S = -k_B \ln 2$. It has to occur in contact with a bath at temperature T , and the second law for the memory gives

$$\Sigma = \Delta S + \left(-\frac{Q}{T}\right) \geq 0 \Rightarrow Q_{\text{erase}} \leq T \Delta S = -k_B T \ln 2.$$

Since the energy of the memory does not change, $\Delta E = 0 = W + Q$, so $W = -Q$ and

$$W \geq k_B T \ln 2 \quad (\text{minimum cost}).$$

3.5 Jarzynski equality

Consider a system with Hamiltonian $H(z, \lambda)$, $z = (q, p)$, initially in equilibrium with a bath at temperature T ,

$$\rho(z) = \frac{e^{-\beta H(z, \lambda_0)}}{Z(\lambda_0)}.$$

The system is then isolated and a protocol changes $\lambda_0 \rightarrow \lambda_1$. The microstate evolves deterministically, $z_{\text{fin}} = \phi^\tau(z_{\text{in}})$, and the work done is

$$W = H(z_{\text{fin}}, \lambda_1) - H(z_{\text{in}}, \lambda_0),$$

which is a random variable (it depends on the initial microstate). Jarzynski's result is the *equality*

$$\langle e^{-\beta W} \rangle = e^{-\beta \Delta F}, \quad \Delta F = -k_B T \ln Z(\lambda_1) + k_B T \ln Z(\lambda_0),$$

whereas the second law is an inequality. Indeed, by Jensen's inequality $\langle e^{-x} \rangle \geq e^{-\langle x \rangle}$,

$$e^{-\beta \Delta F} = \langle e^{-\beta W} \rangle \geq e^{-\beta \langle W \rangle} \Rightarrow \langle W \rangle \geq \Delta F.$$

Proof. Let $\lambda(t)$ go from $\lambda_0 = \lambda(t_0)$ to $\lambda_1 = \lambda(t_1)$. Along a trajectory (on-shell), $\frac{dH}{dt} = \frac{\partial H}{\partial \lambda} \dot{\lambda}$, so

$$W = \int_{t_0}^{t_1} dt \frac{dH}{dt} = \int_{t_0}^{t_1} \frac{\partial H}{\partial \lambda} \dot{\lambda} dt.$$

By Liouville's theorem the phase-space density is conserved along trajectories and the phase-space volume is preserved, $dz_t = dz_0$:

$$\rho(z_t, t) = \rho(z_0, t_0) = \frac{e^{-\beta H(z_0, \lambda_0)}}{Z(\lambda_0)}.$$

Hence

$$\rho(z_t, t) e^{-\beta W} = \frac{e^{-\beta H(z_0, \lambda_0)}}{Z(\lambda_0)} e^{-\beta [H(z_t, \lambda_1) - H(z_0, \lambda_0)]} = \frac{e^{-\beta H(z_t, \lambda_1)}}{Z(\lambda_0)},$$

and integrating over z_t ,

$$\langle e^{-\beta W} \rangle = \frac{Z(\lambda_1)}{Z(\lambda_0)} = e^{-\beta \Delta F}.$$

4 Markov processes and the master equation

The basic setting: an external agent can alter the parameters of the Hamiltonian, while the system is in contact with one or more reservoirs.

4.1 Stochastic processes

A **stochastic process** is a random variable $X(t)$ that depends on a parameter, the time. We write

$$\text{Prob}[x < X(t) < x + dx] = p(x, t) dx.$$

By Kolmogorov's theorem, the process is characterized by the hierarchy of joint probabilities

$$p^{(1)}(x_1, t_1) \geq 0, \quad p^{(2)}(x_2, t_2; x_1, t_1) \geq 0, \quad \dots, \quad p^{(n)}(x_n, t_n; \dots; x_1, t_1) \geq 0,$$

with $t_1 < t_2 < \dots < t_n$, $\int dx p^{(1)}(x, t) = 1$, and the consistency relations (marginalization)

$$p^{(n-1)}(x_{n-1}, \dots, x_1) = \int dx_n p^{(n)}(x_n, x_{n-1}, \dots, x_1).$$

With conditional probabilities,

$$p^{(n)}(x_n, t_n; \dots; x_1, t_1) = p(x_n, t_n | x_{n-1}, t_{n-1}; \dots; x_1, t_1) p^{(n-1)}(x_{n-1}, t_{n-1}; \dots; x_1, t_1).$$

Definition. A process is **Markovian** if

$$p(x_n, t_n | x_{n-1}, t_{n-1}; \dots; x_1, t_1) = p(x_n, t_n | x_{n-1}, t_{n-1}).$$

Hamiltonian evolution is Markovian: the equations of motion can be discretized and only the state at the previous time matters (unless the equations of motion contain an integral over the past). For a Markov process

$$\begin{aligned} p^{(2)}(x_2, t_2; x_1, t_1) &= p(x_2, t_2 | x_1, t_1) p^{(1)}(x_1, t_1), \\ p^{(3)}(x_3, t_3; x_2, t_2; x_1, t_1) &= p(x_3, t_3 | x_2, t_2) p(x_2, t_2 | x_1, t_1) p^{(1)}(x_1, t_1), \end{aligned}$$

and so on. Therefore we only need to know the initial condition $p^{(1)}(x_1, t_1)$ and the transition probability $p(x, t | x', t')$ with $t > t'$.

Chapman–Kolmogorov equation. Integrating the three-point relation over x_2 ,

$$p^{(2)}(x_3, t_3; x_1, t_1) = p^{(1)}(x_1, t_1) \int dx_2 p(x_3, t_3 | x_2, t_2) p(x_2, t_2 | x_1, t_1),$$

so

$$p(x_3, t_3 | x_1, t_1) = \int dx_2 p(x_3, t_3 | x_2, t_2) p(x_2, t_2 | x_1, t_1).$$

4.2 Master equation

We restrict to processes that are homogeneous in time,

$$p(x, t | x', t') = p(x | x'; \tau), \quad \tau = t - t',$$

which do not depend on when the clock is started. *Hypothesis:* for $\tau \ll 1$,

$$p(x | x'; \tau) = (1 - \tau a_0(x')) \delta(x - x') + \tau W(x | x') + \mathcal{O}(\tau^2),$$

where $W(x | x')$ is the **transition rate**. Normalization, $\int dx p(x | x'; \tau) = 1$, gives

$$1 = 1 - \tau a_0 + \tau \int dx W(x | x') + \mathcal{O}(\tau^2) \Rightarrow a_0(x') = \int dx W(x | x').$$

Now take $t_3 = t + \varepsilon$, $t_2 = t$, $t_1 = t' < t$ in the Chapman–Kolmogorov equation:

$$\begin{aligned} p(x_3, t + \varepsilon | x_1, t') &= \int dx_2 \left[(1 - \varepsilon a_0(x_2)) \delta(x_3 - x_2) + \varepsilon W(x_3 | x_2) + \mathcal{O}(\varepsilon^2) \right] p(x_2, t | x_1, t') \\ &= (1 - \varepsilon a_0(x_3)) p(x_3, t | x_1, t') + \varepsilon \int dx_2 W(x_3 | x_2) p(x_2, t | x_1, t') + \mathcal{O}(\varepsilon^2). \end{aligned}$$

Dividing by ε and letting $\varepsilon \rightarrow 0$, with $a_0(x_3) = \int dx_2 W(x_2 | x_3)$,

$$\frac{\partial}{\partial t} p(x_3, t | x_1, t') = \int dx_2 [W(x_3 | x_2) p(x_2, t | x_1, t') - W(x_2 | x_3) p(x_3, t | x_1, t')].$$

Multiplying by $p^{(1)}(x_1, t')$ and integrating over x_1 (dropping the superscript) gives the **master equation**

$$\partial_t p(x, t) = \int dx' [W(x | x') p(x', t) - W(x' | x) p(x, t)]. \quad (4.1)$$

For discrete states,

$$\frac{dp_i}{dt} = \sum_j [W(i | j) p_j - W(j | i) p_i],$$

where p_i is the probability of state i and $W(i | j)$ the transition rate from j to i ; the term $i = j$ does not contribute.

4.3 Detailed balance

The transitions, and therefore the rates $W(x|x')$, are due to the coupling of the system to the external medium. If the only external medium is a bath at temperature $T = 1/(k_B\beta)$, a system with discrete states i of energy ε_i must relax to

$$p_i^{\text{eq}} = \frac{e^{-\beta\varepsilon_i}}{Z}.$$

Stationarity requires

$$0 = \sum_j \left[W(i|j) \frac{e^{-\beta\varepsilon_j}}{Z} - W(j|i) \frac{e^{-\beta\varepsilon_i}}{Z} \right].$$

If each term vanishes separately,

$$\frac{W(i|j)}{W(j|i)} = e^{-\beta(\varepsilon_i - \varepsilon_j)}.$$

This condition on the rates is **detailed balance**: going up in energy is suppressed by the Boltzmann factor relative to going down.

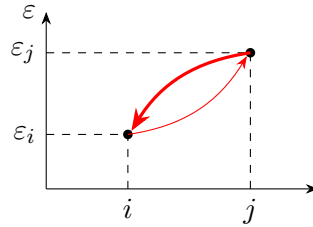


Figure 4.1: Detailed balance: transitions downhill in energy are favoured.

Grand-canonical case. If the environment has temperature T and chemical potential μ , and state i has energy ε_i and n_i particles, the equilibrium state is

$$p_i^{\text{eq}} = \frac{e^{-\beta(\varepsilon_i - \mu n_i)}}{\Theta(T, \mu)},$$

with Θ the grand partition function (e.g. a single level that can be empty, $n_1 = 0$, $\varepsilon_1 = 0$, or occupied, $n_2 = 1$, $\varepsilon_2 = \varepsilon$). Detailed balance reads

$$\frac{W(i|j)}{W(j|i)} = e^{-\beta[\varepsilon_i - \varepsilon_j - \mu(n_i - n_j)]},$$

and then $p(t) \rightarrow p^{\text{eq}}$ as $t \rightarrow \infty$ (for an ergodic network of transitions, i.e. one in which every state can be reached from every other). This is the only condition on $W(i|j)$; the kinetics remains to be specified.

4.4 Thermodynamics of the master equation

We model the action of an external agent through a parameter $\lambda(t)$ that determines the energies of the states, $\varepsilon_i \rightarrow \varepsilon_i(\lambda(t))$. The thermodynamic quantities are

$$E(t) = \sum_i \varepsilon_i(\lambda(t)) p_i(t), \quad N(t) = \sum_i n_i p_i(t), \quad S(t) = -k_B \sum_i p_i(t) \ln p_i(t).$$

It is convenient to write the master equation as $\frac{dp_i}{dt} = \sum_j W_{ij} p_j$ with

$$W_{ij} = \begin{cases} W(i|j), & i \neq j, \\ -\sum_{k \neq i} W(k|i), & i = j, \end{cases} \quad \text{so that} \quad \sum_i W_{ij} = 0.$$

First law.

$$\frac{dE}{dt} = \dot{\lambda} \sum_i p_i \frac{\partial \varepsilon_i}{\partial \lambda} + \sum_i (\varepsilon_i - \mu n_i) \frac{dp_i}{dt} + \mu \sum_i n_i \frac{dp_i}{dt} = \dot{W}_{\text{mech}} + \dot{Q} + \dot{W}_{\text{chem}},$$

with the mechanical work $\dot{W}_{\text{mech}} = \dot{\lambda} \langle \partial \varepsilon / \partial \lambda \rangle$, the heat $\dot{Q} = \sum_i (\varepsilon_i - \mu n_i) \dot{p}_i$ and the chemical work $\dot{W}_{\text{chem}} = \mu \dot{N}$. This agrees with thermodynamics: for system plus environment, $\Delta E + \Delta E_{\text{env}} = \dot{W}_{\text{mech}}$, with $\Delta N + \Delta N_{\text{env}} = 0$ and $\Delta E_{\text{env}} = T \Delta S_{\text{env}} + \mu \Delta N_{\text{env}}$ (no volume change), $-Q = T \Delta S_{\text{env}}$. Hence

$$\begin{aligned} \Delta E &= W_{\text{mech}} + Q + \mu \Delta N \quad (\text{first law; } \mu \Delta N \text{ is the chemical work}), \\ \Sigma &= \Delta S + \left(-\frac{Q}{T} \right) \geq 0 \quad (\text{second law}). \end{aligned}$$

Second law. Since $\sum_i \dot{p}_i = 0$,

$$\frac{dS}{dt} = -k_B \sum_i \frac{dp_i}{dt} \ln p_i = -k_B \sum_{i,j} W_{ij} p_j \ln p_i.$$

Using $\sum_i W_{ij} = 0$, i.e. $\sum_{i,j} W_{ij} p_j \ln p_j = 0$,

$$\frac{dS}{dt} = -k_B \sum_{i,j} W_{ij} p_j \ln \frac{p_i}{p_j} = \frac{k_B}{2} \sum_{i,j} (W_{ij} p_j - W_{ji} p_i) \ln \frac{p_j}{p_i},$$

where in the last step we symmetrized by exchanging $i \leftrightarrow j$. Define the net transition rate (probability current) $J_{ij} := W_{ij} p_j - W_{ji} p_i = -J_{ji}$, and split the logarithm:

$$\frac{dS}{dt} = \frac{k_B}{2} \sum_{i,j} J_{ij} \ln \frac{W_{ij} p_j}{W_{ji} p_i} + \frac{k_B}{2} \sum_{i,j} J_{ij} \ln \frac{W_{ji}}{W_{ij}}.$$

By detailed balance, $\ln(W_{ji}/W_{ij}) = -\beta[(\varepsilon_j - \mu n_j) - (\varepsilon_i - \mu n_i)]$, so the second term is

$$-\frac{1}{2T} \sum_{i,j} J_{ij} [(\varepsilon_j - \mu n_j) - (\varepsilon_i - \mu n_i)] = \frac{1}{T} \sum_{i,j} J_{ij} (\varepsilon_i - \mu n_i) = \frac{1}{T} \sum_i \frac{dp_i}{dt} = \frac{\dot{Q}}{T}.$$

Therefore

$$\frac{dS}{dt} = \dot{\Sigma} + \frac{\dot{Q}}{T}, \quad \dot{\Sigma} = \frac{k_B}{2} \sum_{i,j} (W_{ij} p_j - W_{ji} p_i) \ln \frac{W_{ij} p_j}{W_{ji} p_i} \geq 0,$$

since every term is non-negative: if $W_{ij} p_j - W_{ji} p_i > 0$ the logarithm is positive, and if it is negative so is the logarithm. This is the second law, $\dot{\Sigma} = dS/dt + (-\dot{Q}/T) \geq 0$, for the Shannon entropy.

Quasistatic processes. If $\lambda(t)$ is changed so slowly that at all times

$$p_i(t) = \frac{e^{-\beta[\varepsilon_i(\lambda(t)) - \mu n_i]}}{\Theta(T, \mu, \lambda(t))} = p_i^{\text{eq}}(t),$$

then $\dot{\Sigma} = 0$ during the whole process. If $p_i(t) = p_i^{\text{eq}}(t) + \delta q_i(t)$, then $\dot{\Sigma}$ is quadratic in the perturbation.

4.5 Several reservoirs: non-equilibrium steady states

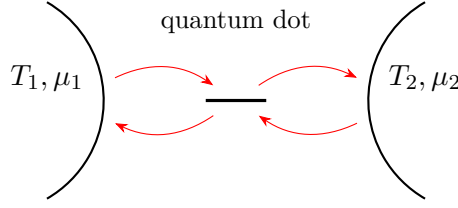


Figure 4.2: A quantum dot coupled to two reservoirs.

If the system is coupled to several reservoirs ν (e.g. a quantum dot between two leads at T_1, μ_1 and T_2, μ_2), the rates are additive,

$$W(i|j) = \sum_{\nu} W^{\nu}(i|j), \quad \frac{W^{\nu}(i|j; \lambda)}{W^{\nu}(j|i; \lambda)} = e^{-\beta_{\nu}[\varepsilon_i - \varepsilon_j - \mu_{\nu}(n_i - n_j)]},$$

each satisfying detailed balance with its own reservoir. With two reservoirs the system does not reach equilibrium, but it can become stationary: the stationary solution $\dot{p}_i = 0$ is a **non-equilibrium steady state** (NESS). The energy balance becomes

$$\frac{dE}{dt} = \dot{\lambda} \left\langle \frac{\partial \varepsilon}{\partial \lambda} \right\rangle + \underbrace{\sum_{\nu} \mu_{\nu} \underbrace{\sum_{i,j} n_i W_{ij}^{\nu} p_j}_{\dot{N}^{\nu}}}_{\dot{W}_{\text{chem}}} + \underbrace{\sum_{\nu} \sum_{i,j} (\varepsilon_i - \mu_{\nu} n_i) W_{ij}^{\nu} p_j}_{\dot{Q}^{\nu}}_{\dot{Q}},$$

and the second law

$$\dot{\Sigma} = \frac{dS}{dt} - \sum_{\nu} \frac{\dot{Q}^{\nu}}{T_{\nu}} \geq 0.$$

5 Stochastic thermodynamics and fluctuation theorems

5.1 Thermodynamics along a trajectory

In stochastic thermodynamics the state $i(t)$ of the system is itself a random variable: each realization of the process is a trajectory that jumps between states at random times.

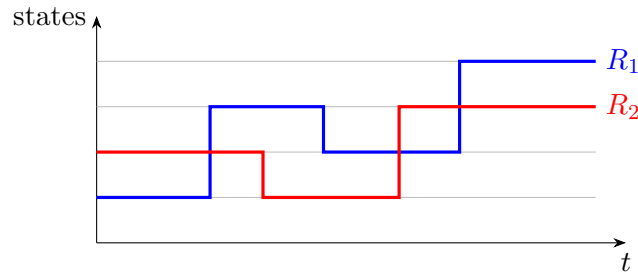


Figure 5.1: Two realizations of the process $i(t)$.

Along a trajectory we define

- the stochastic energy $\varepsilon_{i(t)}(\lambda(t))$,
- the stochastic occupation $n_{i(t)}$,

- the stochastic entropy $s_{i(t)}(t) := -k_B \ln p_{i(t)}(t)$,

whose averages are E , N and S . Writing $\varepsilon_{i(t)} = \sum_j \delta_{i(t),j} \varepsilon_j$, the **first law** along a trajectory is

$$\frac{d}{dt} \varepsilon_{i(t)}(\lambda(t)) = \underbrace{\dot{\lambda} \frac{\partial \varepsilon_{i(t)}}{\partial \lambda}}_{\dot{w}^{\text{mech}}} + \underbrace{\sum_j \left(\frac{d}{dt} \delta_{i(t),j} \right) (\varepsilon_j - \mu n_j)}_{\dot{q}} + \underbrace{\sum_j \mu \left(\frac{d}{dt} \delta_{i(t),j} \right) n_j}_{\dot{w}^{\text{chem}}} :$$

work is done while the system sits in a state and λ changes, heat is exchanged when the system jumps.

Trajectories. A forward trajectory Γ starts in i_0 and jumps $i_0 \rightarrow i_1 \rightarrow i_2 \rightarrow \dots \rightarrow i_n$ at times $t_1, t_2, \dots, t_n$, under the protocol $\lambda(t)$, $0 \leq t \leq t_f$. Along it,

$$W[\Gamma] = \int_0^{t_1} dt \dot{\lambda} \frac{\partial \varepsilon_{i_0}}{\partial \lambda} \Big|_{\lambda(t)} + \int_{t_1}^{t_2} dt \dot{\lambda} \frac{\partial \varepsilon_{i_1}}{\partial \lambda} \Big|_{\lambda(t)} + \dots,$$

$$Q[\Gamma] = \sum_k [\varepsilon_{i_k}(\lambda(t_k)) - \varepsilon_{i_{k-1}}(\lambda(t_k)) - \mu(n_{i_k} - n_{i_{k-1}})].$$

The probability of the trajectory is

$$\text{Prob}[\Gamma] = W_{i_n i_{n-1}}(t_n) \cdots W_{i_2 i_1}(t_2) W_{i_1 i_0}(t_1) p_{i_0}(0)$$

(up to the waiting-time factors and the measure $dt_1 \cdots dt_n$, which are the same for the forward and backward trajectories and cancel in the ratios below).

Definition. The **entropy production of a trajectory** is

$$\Sigma[\Gamma] := k_B \ln \frac{\text{Prob}[\Gamma]}{\text{Prob}[\tilde{\Gamma}]},$$

where $\tilde{\Gamma}$ is the time-reversed trajectory, $i_n \rightarrow i_{n-1} \rightarrow \dots \rightarrow i_0$ at times $t_f - t_n, \dots, t_f - t_1$, generated by the reversed protocol $\tilde{\lambda}(\tilde{t}) = \lambda(t_f - \tilde{t})$ and starting from the final distribution of the forward process.

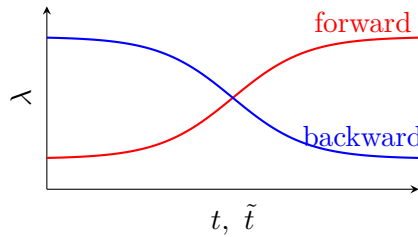


Figure 5.2: Forward protocol $\lambda(t)$ and backward protocol $\tilde{\lambda}(\tilde{t}) = \lambda(t_f - \tilde{t})$.

The probability of the reversed trajectory is

$$\tilde{\text{Prob}}[\tilde{\Gamma}] = W_{i_0 i_1}(t_1) \cdots W_{i_{n-1} i_n}(t_n) p_{i_n}(t_f),$$

so

$$\frac{\text{Prob}[\Gamma]}{\tilde{\text{Prob}}[\tilde{\Gamma}]} = \frac{p_{i_0}(0)}{p_{i_n}(t_f)} \frac{W_{i_1 i_0}(t_1)}{W_{i_0 i_1}(t_1)} \cdots \frac{W_{i_n i_{n-1}}(t_n)}{W_{i_{n-1} i_n}(t_n)}.$$

By detailed balance each ratio of rates is $e^{-\beta[\varepsilon_{i_k} - \varepsilon_{i_{k-1}} - \mu(n_{i_k} - n_{i_{k-1}})]}$, i.e. $e^{-\beta q_k}$ with q_k the heat absorbed in the jump. Hence

$$\begin{aligned} \frac{\text{Prob}[\Gamma]}{\tilde{\text{Prob}}[\tilde{\Gamma}]} &= \frac{p_{i_0}(0)}{p_{i_n}(t_f)} e^{-\beta Q[\Gamma]} = \exp \left\{ \frac{1}{k_B} \left[k_B \ln p_{i_0}(0) - k_B \ln p_{i_n}(t_f) - \frac{Q[\Gamma]}{T} \right] \right\} \\ &= \exp \left\{ \frac{1}{k_B} \left[\Delta s - \frac{Q[\Gamma]}{T} \right] \right\}, \end{aligned}$$

and the entropy production of the trajectory is the change in stochastic entropy minus the heat over T :

$$\Sigma[\Gamma] = \Delta s - \frac{Q[\Gamma]}{T}.$$

Normalization of $\tilde{\text{Prob}}$ gives

$$\left\langle e^{-\Sigma/k_B} \right\rangle = \sum_{\Gamma} \text{Prob}[\Gamma] \frac{\tilde{\text{Prob}}[\tilde{\Gamma}]}{\text{Prob}[\Gamma]} = 1 \Rightarrow \langle \Sigma[\Gamma] \rangle \geq 0.$$

Individual trajectories can have $\Sigma[\Gamma] < 0$; the second law holds on average.

5.2 Fluctuation theorems

The second law states $\Sigma \geq 0$. In stochastic thermodynamics, the thermodynamic entropy production is the average $\Sigma = \langle \Sigma[\Gamma] \rangle$ of a random variable defined on trajectories of microstates (or macrostates) $\Gamma_{\tau,0}$ between times 0 and τ . This random variable satisfies the **detailed fluctuation theorem**

$$\frac{\text{Prob}[\Gamma_{\tau,0}]}{\tilde{\text{Prob}}[\tilde{\Gamma}_{\tau,0}]} = e^{\Sigma[\Gamma_{\tau,0}]/k_B} \Rightarrow \frac{P(\Sigma)}{\tilde{P}(-\Sigma)} = e^{\Sigma/k_B},$$

where the tilde denotes time reversal, and the **integral fluctuation theorem**

$$\left\langle e^{-\Sigma/k_B} \right\rangle = 1.$$

5.3 Measurement and feedback

We now generalize to include measurement and feedback. Example: the Szilard engine. A single particle is in a box in contact with a bath at temperature T . We insert a partition, *measure* that the particle is on the left, and apply *feedback*: we let the piston expand to the right. We extract

$$W = - \int p dV = -k_B T \int_{V/2}^V \frac{dV'}{V'} = -k_B T \ln 2.$$

Here $\Delta F = 0$, and the second law without measurement, $W_d := W - \Delta F \geq 0$ (dissipated work), would require $W \geq 0$. With measurement and feedback the bound becomes

$$\beta(W - \Delta F) \geq -I,$$

and here $I = \ln 2 = H$, since the measurement has no error: $I(X : Y) = H(Y) - H(Y|X)$ with $H(Y|X) = 0$ if $Y = f(X)$ is deterministic. The Szilard engine saturates the bound.

Setting. Markovian stochastic processes without particle exchange: states i with energies $\varepsilon_i(\lambda(t))$, so the grand potential is replaced by the free energy ($\Theta \rightarrow Z$, $\mu = 0$). We want to show the **fluctuation theorem with measurement and feedback**

$$\frac{\text{Prob}[\Gamma_{\tau,0}; \Lambda_\tau]}{\text{Prob}[\tilde{\Gamma}_{\tau,0}; \tilde{\Lambda}_\tau]} = e^{\beta W_d[\Gamma_{\tau,0}; \Lambda_\tau] + I[\Gamma_{\tau,0}; \Lambda_\tau]}.$$

The ingredients:

1. The trajectory $\Gamma_{\tau,0} = \{i_0, i_1, \dots, i_n\}$ with jumps at $t_1, \dots, t_n$, $t_n < \tau$.
2. The protocol Λ_τ : at time t_m we measure a quantity M with possible outcomes m , and from then on the protocol $\lambda_m(t)$ depends on the outcome. The protocol is itself a stochastic process:

$$\lambda(t) = \begin{cases} \lambda_0(t), & 0 < t < t_m, \\ \lambda_m(t), & t_m < t < \tau. \end{cases}$$

3. $\text{Prob}[\Gamma_{\tau,0}; \Lambda_\tau]$ is the joint probability of the trajectory and the protocol.
4. Hypothesis on the measurement: the outcome has probability $p(m|i_m)$, with i_m the state of the system at $t = t_m$.
5. The dissipated work is $W_d[\Gamma_{\tau,0}; \Lambda_\tau] = W - \Delta F$ with $\Delta F = -k_B T \ln Z(\lambda_m(\tau)) + k_B T \ln Z(\lambda_0(0))$. As before,

$$W = \int_0^{t_1} dt \dot{\lambda} \frac{\partial \varepsilon_{i_0}}{\partial \lambda} + \int_{t_1}^{t_2} dt \dot{\lambda} \frac{\partial \varepsilon_{i_1}}{\partial \lambda} + \dots, \quad Q = \sum_k [\varepsilon_{i_k}(\lambda(t_k)) - \varepsilon_{i_{k-1}}(\lambda(t_k))],$$

(with $\lambda = \lambda_m$ after t_m), $\Delta E = \varepsilon_{i_n}(\lambda_m(\tau)) - \varepsilon_{i_0}(\lambda_0(0))$ and $\Delta E - W = Q$.

6. $I[\Gamma_{\tau,0}; \Lambda_\tau]$ is related to the information acquired in the measurement; its form follows from the proof.
7. The reversed protocol and trajectory:

$$\tilde{\lambda}(t) = \begin{cases} \lambda_m(\tau - t), & 0 < t < \tau - t_m, \\ \lambda_0(\tau - t), & \tau - t_m < t < \tau, \end{cases} \quad \tilde{\Gamma}_{\tau,0} = \{i_n, i_{n-1}, \dots, i_0\} \text{ at times } \tau - t_n, \dots, \tau - t_1,$$

so that time intervals are preserved: $(\tau - t_{n-2}) - (\tau - t_{n-1}) = t_{n-1} - t_{n-2}$.

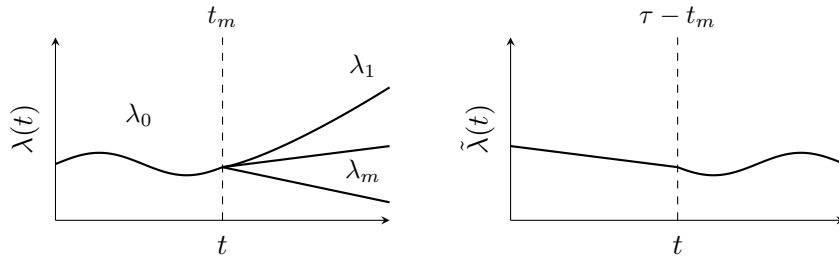


Figure 5.3: Feedback protocol: after the measurement at t_m the protocol branches according to the outcome m (left). The reversed protocol for a given outcome (right).

Proof. The forward probability is

$$\text{Prob}[\Gamma_{\tau,0}; \Lambda_\tau] = p(m|i_m) W_{i_n i_{n-1}}(t_n) \cdots W_{i_2 i_1}(t_2) W_{i_1 i_0}(t_1) p_{i_0}^{\text{eq}}(\lambda_0(0)),$$

with

$$p_{i_0}^{\text{eq}}(\lambda_0(0)) = \frac{e^{-\beta \varepsilon_{i_0}(\lambda_0(0))}}{Z(\lambda_0(0))},$$

where the rates are evaluated at $\lambda(t_k)$. The forward processes generate the outcome m with probability

$$p(m) = \sum_i p(m|i) p_i(t_m),$$

the probability of observing $M = m$ in the ensemble of forward realizations. The reversed process is defined as follows: choose m with probability $p(m)$, use the protocol $\tilde{\lambda}_m(t)$, and start from $p_{i_n}^{\text{eq}}(\lambda_m(\tau))$. Then

$$\text{Prob}[\tilde{\Gamma}_{\tau,0}; \tilde{\Lambda}_\tau] = p(m) W_{i_0 i_1}(\tau - t_1) \cdots W_{i_{n-1} i_n}(\tau - t_n) p_{i_n}^{\text{eq}}(\lambda_m(\tau)),$$

with $\tilde{\lambda}(\tau - t_k) = \lambda(t_k)$. The ratio is

$$\frac{\text{Prob}[\Gamma_{\tau,0}; \Lambda_\tau]}{\text{Prob}[\tilde{\Gamma}_{\tau,0}; \tilde{\Lambda}_\tau]} = \frac{p(m|i_m)}{p(m)} \frac{p_{i_0}^{\text{eq}}(\lambda_0(0))}{p_{i_n}^{\text{eq}}(\lambda_m(\tau))} \frac{W_{i_1 i_0}(\lambda(t_1))}{W_{i_0 i_1}(\lambda(t_1))} \cdots \frac{W_{i_n i_{n-1}}(\lambda(t_n))}{W_{i_{n-1} i_n}(\lambda(t_n))}.$$

The ratio of equilibrium distributions is $e^{\beta[F(\lambda_0(0)) - F(\lambda_m(\tau))] + \beta[\varepsilon_{i_n}(\lambda_m(\tau)) - \varepsilon_{i_0}(\lambda_0(0))]} = e^{\beta(\Delta E - \Delta F)}$, and the product of rate ratios is $e^{-\beta Q[\Gamma_{\tau,0}; \Lambda_\tau]}$. With $\Delta E - Q = W$,

$$\frac{\text{Prob}[\Gamma_{\tau,0}; \Lambda_\tau]}{\text{Prob}[\tilde{\Gamma}_{\tau,0}; \tilde{\Lambda}_\tau]} = e^I e^{\beta(W - \Delta F)} = e^{\beta W_d + I}, \quad I[\Gamma_{\tau,0}; \Lambda_\tau] = \ln \frac{p(m|i_m)}{p(m)} = \ln p(m|i_m) - \ln p(m).$$

Average information. Averaging over protocols and trajectories, $\text{Prob}[\Lambda_\tau]$ $\text{Prob}[\Gamma_{\tau,0}|\Lambda_\tau]$, and summing over all jumps except the state $i = i_m$ at the measurement (the rates sum to the propagated distribution $p_i(t_m)$),

$$\begin{aligned} \langle I \rangle &= \sum_{m,i} p(m|i) p_i(t_m) [\ln p(m|i) - \ln p(m)] = - \sum_i p_i H(M|i) - \sum_m p(m) \ln p(m) \\ &= H(M) - H(M|X), \end{aligned}$$

the mutual information between the measurement outcome and the microstate X . From the integral version, $\langle e^{-\beta W_d - I} \rangle = 1$, and Jensen's inequality gives the generalized second law $\beta \langle W_d \rangle \geq -\langle I \rangle$.